ZAVIONIX / ETHOS

RDTgauge User Manual

A turbine dashboard using dial gauges for RPM, EGT, ECU battery, pump and fuel, with configurable gauge scales.

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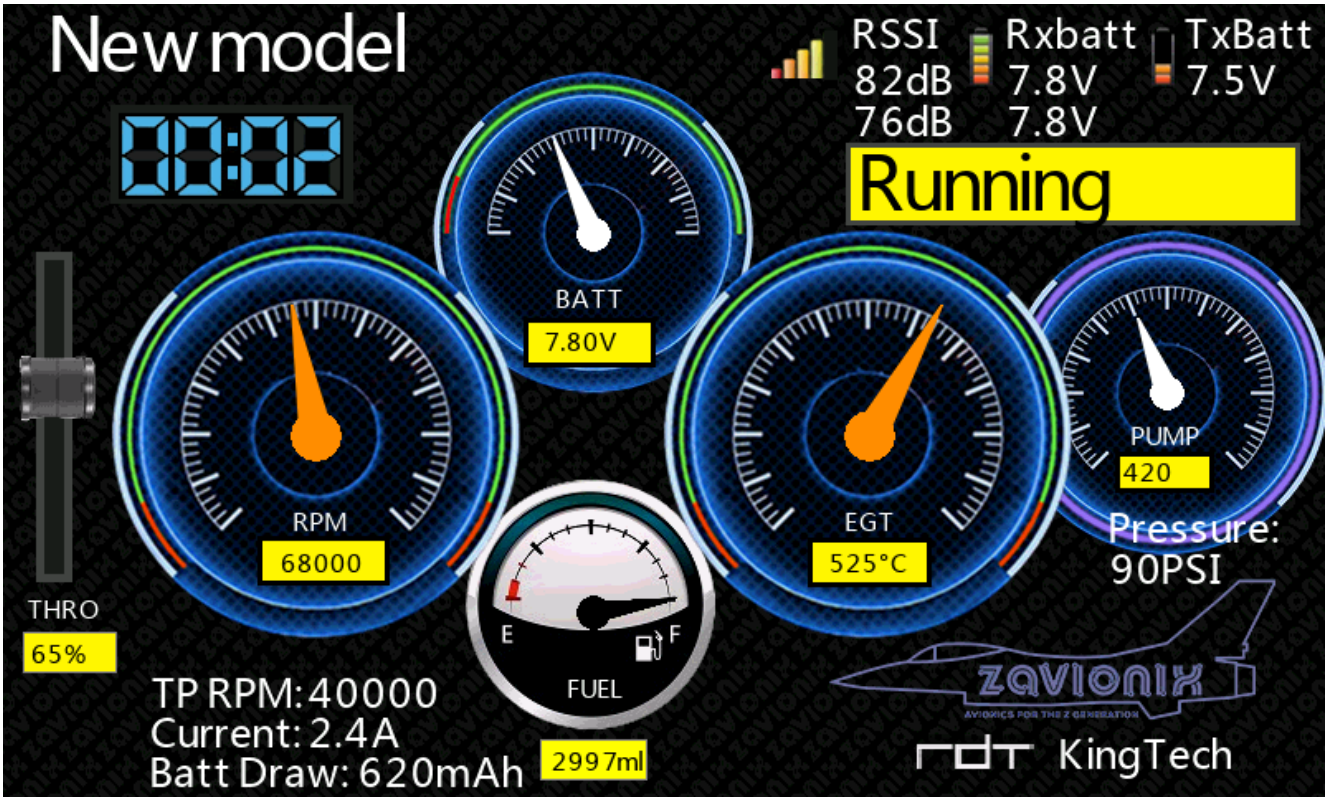


Figure 1. Actual ETHOS simulator rendering with simulated sensor data. The example is not a flight record or a recommended operating setup.

Read this guide in order for first installation. Use the settings and sensor reference when adapting an existing model.

1. Before you begin

A turbine dashboard using dial gauges for RPM, EGT, ECU battery, pump and fuel, with configurable gauge scales.

You need a compatible color-screen ETHOS transmitter or simulator, the complete widget folder, and the relevant telemetry sources already available in the model. These captures use the VS Code-cached X20S ETHOS 26.1.3 simulator engine at 800 x 480. Other screens may arrange or scale the content differently.

This manual covers installation, configuration, every captured widget-specific form field, normal use and troubleshooting. Menu labels are written as they appear in this version. Numeric examples are demonstration settings, not model-specific recommendations.

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2. Install and add the widget

1. Back up the model and existing widget folder. Download the RDTgauge repository ZIP from https://github.com/i3dm/RDTgauge (Code / Download ZIP) or use the appropriate version of the repository.

2. Extract the archive and copy the whole src/RDTgauge folder into the radio storage /scripts directory, producing /scripts/RDTgauge/main.lua. Keep the companion Lua files, images and audio in their supplied relative locations. Do not copy only main.lua.

3. Restart ETHOS or reload Lua through the facilities supported by your firmware. If upgrading, use one complete version of the package rather than mixing old and new module files.

4. Open Configure screens (DISP in the simulator). Add or select a screen and choose a suitable layout; for a full-screen view, long-press its tab and use Change layout. Tap the widget area and select Zavionix RDT gauge from Widget.

5. Open the widget configuration, set the model-specific parameters, and assign telemetry sources. Use the back arrow to return to the dashboard. Reopen the settings and confirm the assignments persist.

Use only the production widget folder on a real transmitter. Documentation sensor fixtures and simulator radio copies are not part of the installation.

3. Configure telemetry and read the display



Figure 2. Widget configuration in ETHOS. Scroll down to reach the remaining settings and source selectors; the reference lists fields below the visible area as well.

Power the intended telemetry hardware in a controlled setup and discover its sensors in the ETHOS model telemetry page. Confirm each value and unit there before assigning it in the widget. Receiver 1/2 and engine left/right inputs must refer to the intended physical source.

A missing value or a No ... label normally means an unassigned or unavailable source. A plausible but incorrect value often means the wrong source, unit or scale. Inspect the source in ETHOS telemetry first; do not compensate for a bad assignment by changing an unrelated capacity or alert setting.

Where displayed, RSSI and RxBatt come from your selected receiver sources; TxBatt comes from the transmitter. The model name and picture come from ETHOS model settings. A missing model image does not imply a telemetry failure. The example photograph is specific to the test model.

4. Operating guide

ECU selection and telemetry scaling

Choose ECU Type before binding sources. Supported menu choices are Jetcat, Projet, Xicoy V6-V10, Kingtech G2-G5, Jetcentral, Swiwin, Xicoy X, Orbit, AMT, Linton V1, Linton V2, Linton V3 and EnJet. Different ECUs reuse telemetry channels for different meanings.

RPM, Temp 1 (EGT), Temp 2 (Status), ADC3 (ECU V) and ADC4 (Pump PW) form the main turbine input set. DIY 1/2/3, current and consumption are protocol-dependent. A field appearing in the form does not mean every ECU provides it. Leave unsupported inputs unassigned.

Compare the widget with the ECU or its ground terminal at stopped, idle and representative running conditions. An RPM source value of 680 is decoded as 68,000 RPM in the KingTech example. Pump source 4.2 produces display value 420 and DIY 1 value 6500 produces 65% throttle in that example. These scaling examples must not be applied to every ECU.

The illustrated KingTech status code 16 decodes as Running, with 525 C EGT and 7.80 V ECU supply. A wrong ECU setting can make both the status text and numerical values look credible but incorrect.

Fuel estimate, reset and calibration

For Xicoy V6-V10, KingTech, Swiwin, Xicoy X, Orbit, AMT and Linton V1/V2/V3, the widget calculates fuel use from pump-related telemetry with ECU-specific factors. For Jetcat, Projet, Jetcentral and EnJet, it uses ECU-reported fuel information. This is not one universal consumed-volume formula.

Enter the usable tank volume and assign a Fuel Reset Switch for calculated fuel. After filling the tank, activate the reset switch, confirm the displayed quantity returns to the configured tank size, then release it. A switch held active can keep restoring the estimate and conceal consumption.

Fuel Factor changes the rate of calculated use. Calibrate by comparing the change in the displayed estimate with a measured volume used under representative running conditions. Increase the factor if indicated use is too small; decrease it if indicated use is too large. Repeat across different power settings because pump-derived estimates need not be linear.

For an ECU-reported fuel channel, configure and reset that measurement at its source according to the ECU setup. Do not assume a widget reset changes the ECU's own fuel counter. Verify full-tank and partial-tank readings independently.

The screenshot begins near a 3000 ml tank and may show a slightly smaller value because the simulated pump is running. It demonstrates the calculation path, not calibrated fuel accuracy.

Fuel and engine-cut alerts

Fuel Alert enables the low remaining-fuel sound. Fuel Alert (ml) defines the reserve threshold; Fuel Alert Rate (s) sets the repeat interval. Verify both the source validity and the estimate before testing the alert.

Engine Cut Alert uses the configured RPM threshold and its own repeat interval. The entry is in hundreds of RPM: 300 is 30,000 RPM, not 300 RPM. It is an audible indication and does not stop or restart an engine.

Keep the supplied audio files with the widget and confirm the transmitter sound level. The headless simulator captures in this manual do not establish audible playback or complete alert transition behavior.

Gauge display

Max RPM(K) and Idle RPM(K) set the RPM dial scale/reference in thousands of RPM. Min Batt(V) and Max Batt(V) define the ECU-voltage scale. Max Pump and Max EGT(C) set the pump and EGT dial ranges.

For example, Max RPM 150 corresponds to 150,000 RPM, while battery limits 6.5 and 10.0 are volts. Choose scales that make the expected operating range readable. These fields do not program ECU limits or provide engine protection.

5. Settings and sensor reference

The following labels were read from the running widget configuration. Example numbers describe the documentation setup or initialized fields; they are not guaranteed factory defaults for a previously saved model. Source assignments and switches must be chosen for your model. Decimal values below use the units displayed in the form.

Parameters

Menu field	Choices / example / range	Purpose
ECU Type	Jetcat; Projet; Xicoy V6-V10; Kingtech G2-G5; Jetcentral; Swiwin; Xicoy X; Orbit; AMT; Linton V1/V2/V3; EnJet. Example: Kingtech G2-G5.	Select the actual ECU family before assigning telemetry. This controls status decoding, scaling and fuel calculations.
Stick Mode	Mode 1; Mode 2; Mode 3; Mode 4	Match your transmitter stick mode so the on-screen controls/trims are placed correctly. This setting does not remap the transmitter controls.
Fuel Factor	Example 100; range 1 to 10000	Calibration multiplier for fuel use. Higher values consume the calculated tank faster. In the twin display, calibrate each engine independently.
Fuel Tank Size (ml)	Example 3000; range 1 to 10000	Usable full-tank volume in ml. The twin display applies the configured tank size to each engine estimate.
Fuel Reset Switch	Unassigned; select a switch	Choose a switch to restore the calculated fuel quantity to the configured tank size after refuelling. Return it inactive so the estimate can decrease.
Fuel Alert	On / Off; example On	Enable or disable the named audible alert. Confirm the associated threshold and repeat interval; disabling an alert does not correct bad telemetry.
Fuel Alert (ml)	Example 1000; range 1 to 10000	Remaining-fuel alert threshold in ml. Set a reserve appropriate to the model and verify the estimate by measurement.
Fuel Alert Rate (s)	Example 5; range 1 to 10	Minimum repeat interval in seconds for the associated alert while its condition persists. The sound files must remain in the widget folder.
Engine Cut Alert	On / Off; example On	Enable or disable the named audible alert. Confirm the associated threshold and repeat interval; disabling an alert does not correct bad telemetry.
Engine Cut RPM (x 100)	Example 300; range 10 to 1500	Engine-cut threshold, entered in hundreds of RPM: 300 means 30,000 RPM. This is an alert threshold, not a throttle or ECU command.
Engine Cut Alert Rate (s)	Example 5; range 1 to 10	Minimum repeat interval in seconds for the associated alert while its condition persists. The sound files must remain in the widget folder.
Max RPM(K)	Example 150; range 1 to 300	Gauge scale/reference setting. RPM values marked K are thousands of RPM. These settings change the display, not ECU operating limits.

Menu field	Choices / example / range	Purpose
Idle RPM(K)	Example 50; range 1 to 100	Gauge scale/reference setting. RPM values marked K are thousands of RPM. These settings change the display, not ECU operating limits.
Min Batt(V)	Example 6.5; range 0.1 to 15	Gauge scale/reference setting. RPM values marked K are thousands of RPM. These settings change the display, not ECU operating limits.
Max Batt(V)	Example 10; range 0.1 to 30	Gauge scale/reference setting. RPM values marked K are thousands of RPM. These settings change the display, not ECU operating limits.
Max Pump	Example 10000; range 1 to 100000	Gauge scale/reference setting. RPM values marked K are thousands of RPM. These settings change the display, not ECU operating limits.
Max EGT(C)	Example 800; range 1 to 1200	Gauge scale/reference setting. RPM values marked K are thousands of RPM. These settings change the display, not ECU operating limits.

Source assignments

Source selector	Required value and interpretation
RPM Sensor	Engine-speed telemetry. The turbine decoder can apply ECU-specific scaling; compare the displayed RPM with the ECU reading.
Temp 1 Sensor (EGT)	Exhaust gas temperature in degrees Celsius. Use the ECU temperature channel, not its status channel.
Temp 2 Sensor (Status)	Numeric ECU status code, decoded using ECU Type. An incorrect ECU choice can produce a misleading status message.
ADC3 Sensor (ECU V)	ECU supply-voltage channel. Follow the selected ECU protocol mapping; do not substitute receiver voltage.
ADC4 Sensor (Pump PW)	Pump telemetry channel. Its units and scaling depend on ECU Type; it also feeds the fuel estimate for several ECU families.
DIY 1 Sensor	ECU-specific auxiliary channel. Its role may be fuel, throttle, turboprop RPM or another auxiliary value; use the ECU mapping in this manual.
DIY 2 Sensor	ECU-specific auxiliary channel. Its role may be fuel, throttle, turboprop RPM or another auxiliary value; use the ECU mapping in this manual.
DIY 3 Sensor	ECU-specific auxiliary channel. Its role may be fuel, throttle, turboprop RPM or another auxiliary value; use the ECU mapping in this manual.
Current Sensor	Measured current in amperes. For a turbine, assign the protocol current channel only when supported by the selected ECU.
Consumption Sensor	ECU consumption channel where supported; the display may show battery draw in mAh. It is distinct from estimated remaining fuel.
RSSI 1 Sensor	Receiver signal-strength source in dB. Keep receiver 1 and receiver 2 distinct; leave an unavailable second link unassigned.

Source selector	Required value and interpretation
RSSI 2 Sensor	Receiver signal-strength source in dB. Keep receiver 1 and receiver 2 distinct; leave an unavailable second link unassigned.
Rxbatt 1 Sensor	Receiver supply voltage in volts. This is separate from transmitter voltage and, where applicable, the main propulsion battery.
Rxbatt 2 Sensor	Receiver supply voltage in volts. This is separate from transmitter voltage and, where applicable, the main propulsion battery.
Air Pressure Sensor	Air-system pressure source. Verify the sensor units against the PSI display before relying on the number.

Telemetry identifiers used by automatic matching

This is a matching aid from RDTcombined's protocol map, not a replacement for verifying the actual ECU telemetry names and units. Hexadecimal app IDs identify channels. Standalone widgets can be assigned manually using the same hardware information.

ECU family	Channel matching notes
Standard main channels	RPM 0x0500; EGT 0x0400; Status 0x0410; ECU V 0x0900; Pump 0x0910. Exceptions below override these.
Jetcat / Projet	DIY 1: 0x0A10; DIY 2: 0x0A20.
Xicoy V6-V10	DIY 1: 0x0A10; current/consumption matching uses 0x0200.
Kingtech G2-G5 / Xicoy X	DIY 1/2/3: 0x0A10 / 0x0A20 / 0x0A30.
Jetcentral / Linton V1 / Linton V2	DIY 1/2/3: 0x0A10 / 0x0A20 / 0x0A30; current/consumption: 0x0200.
Swiwin	ECU voltage name VFAS; pump name Current; current/consumption: 0x0200.
Orbit / AMT	Orbit DIY 2: 0x0A20. AMT DIY 1/2: 0x0A10 / 0x0A20.
Linton V3	DIY 3: 0x0A30; current/consumption: 0x0200.
EnJet	RPM 0x0200; EGT 0x0201; Status 0x0204; ECU V 0x0202; Pump 0x0207; DIY 1 0x0206; current/consumption 0x0203.

Current and consumption may share an app-ID candidate in the matching map; confirm their sensor names, units and values separately. The map does not establish that one selected source correctly supplies both quantities.

6. Troubleshooting and verification

Symptom	Checks
Widget absent from selector	Check /scripts/RDTgauge/main.lua and all companion files. Restart ETHOS and inspect Lua messages. Make sure the archive did not create an extra nested folder.
Error after update	Restore a complete matching package version. Check available Lua memory and the exact error in the simulator/radio log; retain the model backup.

Symptom	Checks
Missing / No sensor value	Inspect Model / Telemetry, power the sensor, rediscover if necessary, then reselect the correct source in the widget.
Value has wrong magnitude	Check units, decimals, ECU selection and upstream sensor scaling against a known reference. Do not change warning thresholds to conceal a scaling error.
Values do not change	Compare the original telemetry page. A retained or stale last value is not proof of an active link. Confirm fresh telemetry before use.
Text is clipped	Use a larger widget area/full-screen layout. Cells also has Font Size. Check the intended screen size and version.
No audible alert	Verify enable switch, threshold, repeat interval, telemetry validity, audio files and transmitter volume. Headless screenshots cannot verify the speaker.
Settings appear different	Confirm the selected model, widget instance and, for combined widgets, Widget Display. Each can have its own saved settings.

Model acceptance check

1. Reopen the saved model and verify every configured source and unit.

2. Compare the widget with the underlying telemetry at two or more known values. For twin/battery pairs, vary one side at a time.

3. Check resets, timer controls and any alert conditions relevant to your model in a controlled test. Observe the physical transmitter audio where alerts matter.

4. Verify behavior when a source becomes unavailable and after restoring it. Do not infer source-loss behavior from a static screenshot.

5. Record the final capacities, thresholds, scaling and calibration values with the model backup.

7. Screenshot provenance and version record

The screenshots were produced by the genuine ETHOS X20S 26.1.3 engine (a074fdfd) cached by the VS Code ETHOS extension, driven headlessly through the ethos-tools PR #2 runner. This validates Lua execution and rendering in that engine; it is not a claim that the VS Code webview or every supported transmitter was tested.

Documentation-only copies supplied genuine ETHOS sensor objects and assigned them to the widget. The rendering implementation was retained. Most inputs used DIY sensors refreshed with rawValue; Cells used injected S.Port cell frames to provide an actual multi-cell source. Demonstration settings were initialized independently of the copied model. These fixtures were not installed in the production widget repository.

The simulation method was informed by WingFlight's simulator sensor implementation. The example values are synthetic, not flight measurements. Static captures and configuration inspection do not establish calibrated measurement accuracy, full alert-state coverage, audible playback or physical control performance.

Widget: RDTgauge 3.1.12. Source commit inspected: e65157bfdb0439018ed9cb4cfb7f5ff3f7e307c3. Documentation edition: 15 October 2026. Repository: https://github.com/i3dm/RDTgauge.

Source links

Widget source and updates

ETHOS simulator automation PR

WingFlight simulator sensor approach